Jana Sunuwaai

AI-Powered Civic Reporting

जन सुनुवाई — *People's Voice*

"From photo to government desk in under 60 seconds."

Reports

Community submissions across Kathmandu Valley

Resolution Rate

Verified closures after routing and follow-up

<3s AI Analysis

Rapid classification and routing per image

Technologies: YOLOv8 · FastAPI · React 18 · PostGIS · Framer Motion

THE PROBLEM

Kathmandu Has a Civic Crisis Nobody Is Solving

"Over 40% of Kathmandu's roads are damaged — yet most go unreported and unresolved."

No Reporting Channel

Citizens lack a frictionless, mobile-first way to report potholes, broken lights, and drainage failures.

Zero Accountability

Reports often disappear — no tracking, status updates, or clear ownership.

Wrong Departments

Manual routing sends issues to incorrect teams, adding weeks of delay.

THE SOLUTION

Meet Jana Sunuwaai

जन सुनुवाई | *People's Voice*

Full-stack, AI-powered civic reporting: citizens photograph an issue, pin GPS, and the platform classifies and routes to the right government desk in under 60 seconds.

> "Every pothole reported is a step toward a better city."

Hyperlocal

Built for Nepal's wards and road typologies — fast localisation and mapping.

AI-Powered

YOLOv8 detects severity, size, and road type in under 3 seconds.

Community-Driven

Citizens upvote and verify reports to improve triage and trust.

HOW IT WORKS

Four Steps. Under 60 Seconds.

1 · Choose Category

Road, Electrical, Water, Waste, or Public Spaces — clear, local categories for fast routing.

2 · Take a Photo

AI-optimised capture guidance ensures usable images (direct shots, clear context).

3 · Share GPS

Automatic GPS or manual drag-pin — geo-accuracy to ~8 metres for precise handoff.

4 · Submit & Relax

AI analyses, classifies, and routes instantly; flagged low-confidence items go to human review.

Detections below 60% confidence are flagged for human review before routing.

THE AI ENGINE

What Our AI Detects

YOLOv8 — trained on Nepal road imagery

< 3 SECONDS PER IMAGE

- Detection Confidence
- Severity (Low / Medium / High / Critical)
- Estimated Width (cm)
- Estimated Depth (cm)
- Surface Area (sqm)
- Road Type
- Surface Material
- Weather at Report Time

TECHNOLOGY

Built on Production-Grade Technology

BACKEND & AI

Orange border

- FastAPI (Python) — high-performance API
- PostgreSQL + PostGIS — geo queries and storage
- Redis + Celery — async workers and task queue
- YOLOv8 — production-ready object detection
- Alembic — reliable schema migrations

FRONTEND

Blue border

- React 18 + Vite — snappy UX
- Tailwind CSS — consistent styling
- Framer Motion — smooth interactions
- Leaflet / MapLibre — interactive maps
- Zustand — lightweight state management

Architecture Note

AI pipeline runs as a decoupled async Celery worker so the API remains responsive under heavy load.

TRACTION & IMPACT

Built. Tested. Real.

1,240 Reports Logged

Validation Highlights

- YOLOv8 trained on Nepal road imagery with multiple validation runs
- Full-stack application: API, frontend, and AI pipeline integrated in staging
- Geo-aware clustering and live map with ~8m GPS accuracy
- Community verification with upvoting to prioritise high-impact issues
- Smart routing with severity triage and human fallback for low-confidence cases

WHY US

The Right Team. The Right Problem. The Right Time.

Why This Problem?

Infrastructure repairs are urgent; early fixes are ~10× cheaper than major reconstructions. No Nepal-focused civic platform exists at scale.

Why This Solution?

AI routing removes manual triage, mobile-first UX reduces friction, and localisation ensures relevant routing per ward.

Why This Team?

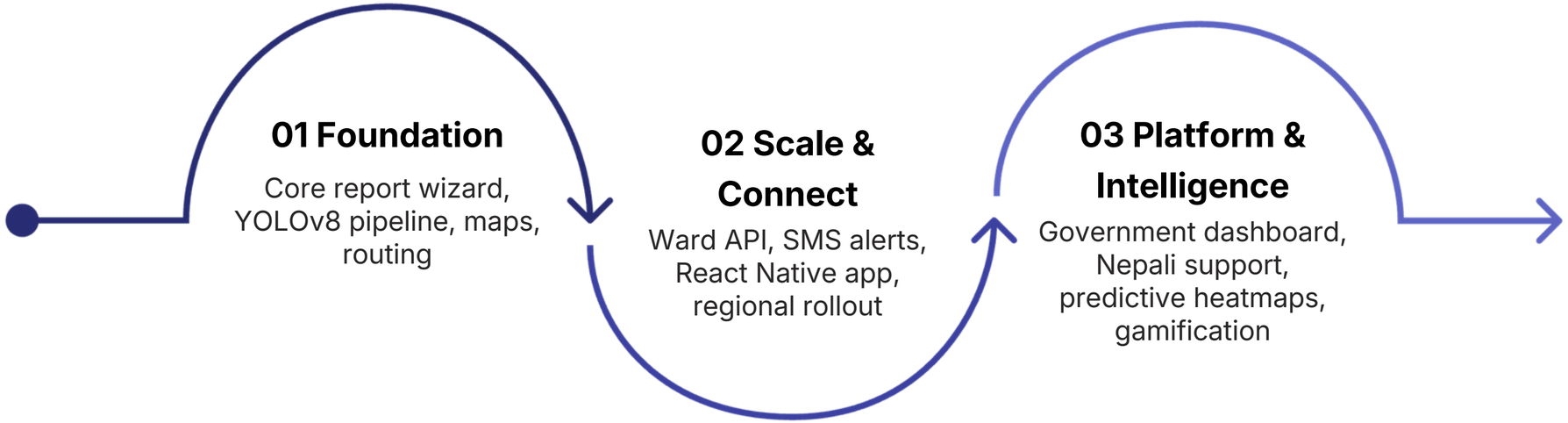
Backend Lead (FastAPI/PostGIS), AI Engineer (YOLOv8/TTA), Frontend Lead (React 18) — complementary production experience.

Market Gap

SeeClickFix (US) and FixMyStreet (UK) serve other markets — Nepal lacks a dedicated platform; Jana Sunuwaai fills that gap.

ROADMAP

What's Next



1

Phase 01 — Foundation

Core report wizard · YOLOv8 AI pipeline · Community feed & maps · Department routing

1

Phase 02 — Scale & Connect

Ward API integration · SMS notifications · React Native app · Regional rollout

1

Phase 03 — Platform & Intelligence

Government dashboard · Nepali language support · Predictive heatmaps · Gamification

CALL TO ACTION

How Municipal Partners Can Join

1

Integration

Provide ward boundary & department endpoint details for seamless routing.

2

Pilots

Run a 3-month pilot in selected wards to measure impact and cost savings.

3

Funding & Support

Partner with civic tech donors to scale infrastructure and SMS outreach.

Primary contact: Program Lead — jana@sunuwai.org · Demo & API docs available on request.

Next Steps

Schedule a demo · Share ward API credentials · Launch pilot within 6 weeks